End Semester Examination, April – May, 2025

Degree: **B.Tech**, Semester: **Second**

Stream: **Biotech/CSE(AI)/CSE(AI & ML)/ CSE(AIML)/CSE(Cyber Security) /CSE(Data Science)/CSE(IOT) /CSE(IOT, CS, BT)/ CSE(IOTCSBT)/CSE(Networks) /Robotics and AI/CSE/CSIT/CST/IT/CSBS/ECE/ME/EE/EEE**

Paper Code: **ESCME202**

Paper Name: **Engineering Mechanics - II**

Full Marks: **100** Duration: **3 Hours**

Part - A
Attempt any 10 questions out of 15 questions
Each question carries 2 Marks (2 × 10)

1. What is the force of gravity? How does it affect the body, when **2**
i) it is allowed to fall downwards
ii) it is projected upwards

2. What is meant by variable acceleration? Give one example. **2**

3. What is the difference between rectangular (Cartesian) and polar coordinates? **2**

4. A 10 kg object is moving at a velocity of 5 m/s when a constant force of 30 N is applied for 4 seconds. Calculate the final velocity of the object. **2**

5. How does D'Alembert's Principle transform a dynamic problem into a static one? **2**

6. Why does a gun recoil when fired? **2**

7. State the principle of conservation of energy and explain why it holds true in an isolated system. **2**

8. Distinguish between elastic and inelastic collisions in terms of energy conservation. **2**

9. What are the units of work done? What is the relation between work and power? **2**

10. When does a car skid and when does it overturn on a level circular road? **2**

11. What is damping and its source? **2**

12. List out methods of finding the natural frequency of free longitudinal vibration. **2**

13. How would you locate the instantaneous centre of a rigid link moving with the combined motion of rotation and translation? **2**

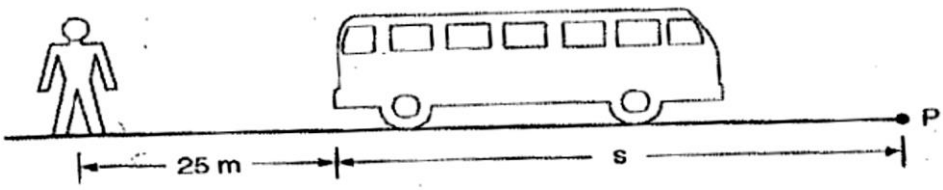
14. Explain the significance of 'natural frequency in a vibrating system. **2**

15. What are the factors the moment of inertia depends on? **2**

Part - B

Attempt any 6 questions out of 9 questions
Each question carries 5 Marks (5 × 6)

16. A bus started moving with an acceleration of 0.5 m/sec². A man standing 25 m behind a bus started running at constant speed of 4.5 m/sec after just starting the bus. After what time will the man catch the bus? **5**

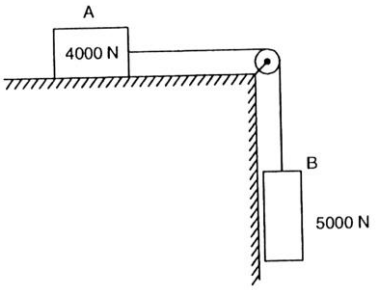


17. Explain the procedure adopted for obtaining the relations for acceleration of two bodies and tension in the string when one of the bodies is hanging free and the other is lying on **5**
    i) smooth inclined plane.
    ii) rough inclined plane.

18. Derive the relationship between the linear motion of the geometric centre and the angular motion of a wheel rolling without slipping. **5**

19. A train of 5 MN weight is moving up a plane whose gradient is 1 in 100. When the train is moving with a speed of 20 km/hr, the acceleration is $0.2 m/sec^2$. Frictional resistance is 5 N per kN weight of the train. Determine the power developed by the engine. **5**

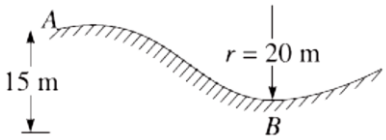
20. Two blocks A and B of weight 4000 N and 5000 N respectively are connected by an inextensible cable as shown in the figure. If the system is released from rest, find the velocity of the block A after it has moved 5 m. Assume the pulley is frictionless and weightless and the coefficient of friction between the block and the horizontal surface is 0.3. **5**



21. A simple pendulum of amplitude 40 performs 24 oscillations in one minute. Find **5**

a) Length of the pendulum.
b) Maximum acceleration of the bob.
c) Maximum linear velocity of the bob.
d) Maximum angular velocity of the bob.

22. A roller weighing 1 kN starts rolling down from position A on a smooth surface as shown in Figure. What will be its velocity at B and the vertical reaction on it from the surface? **5**



23. A ball is projected from the ground at an angle of 60° with the horizontal with a velocity of 30 m/sec. Find the distance of the ball from the point of projection after 2 seconds and its height above the ground. **5**

24. Describe the phenomenon of combined motion of rotation and translation. Give few examples. **5**

Part - C
Attempt any 5 questions out of 8 questions
Each question carries 10 Marks (10 × 5)

25. The brakes are applied to a car of mass 1000 N travelling at 172 km/hr. Determine the time required to stop the car **10**

i) on dry road, $\mu = 0.5$.

ii) on a wet road, $\mu = 0.1$.

26. A block of weight 80 N is hung from a string which is wrapped around a smooth pulley of radius 15 cm and 120 N weight. The system starts from rest, and the block sinks down through 20 cm and spring extends by 2cm. Find the velocity of the block. Take the radius of gyration of the pulley = 10cm. **10**

27. A football is kicked with an initial velocity of 30 m/s at an angle of 45° from the ground. **10**

i) Find the time taken to reach the highest point.

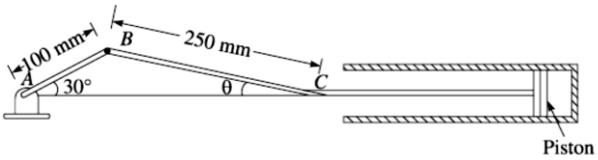
ii) Determine the maximum height reached.

iii) Calculate the total time of flight.

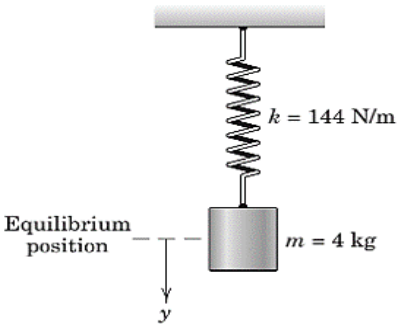
iv) Find the horizontal range.

v) If the ball is kicked on a hill that is 5 m above the ground, how far will it land from the initial point?

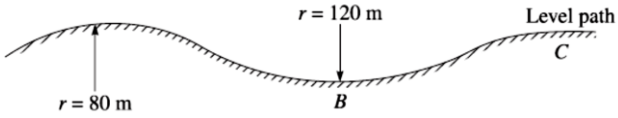
28. Length of crank AB is 100 mm and that of connecting rod is 250 mm as shown in Figure. If the crank is rotating in clockwise direction at 1500 rpm, for the position shown in Figure, determine the angular velocity of the connecting rod and the velocity of the piston. **10**



29. For the spring-mass system shown, determine the static defection the system period and the maximum velocity which result if the cylinder is displaced 0.1 m downward from its equilibrium position and released. **10**



30. An automobile weighing 25 kN moves on a road, the longitudinal section of which is shown in Figure. If it moves with uniform velocity of 50 kmph, what is the vertical reaction experienced at points A, B and C. **10**



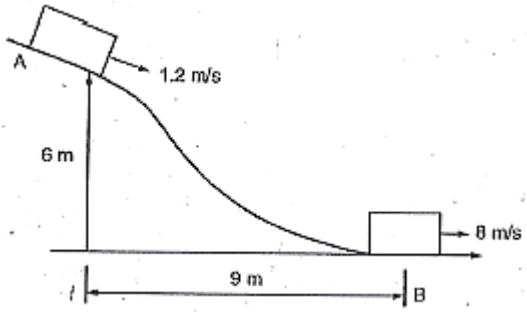
31. A mass of 280 kg falls from a height of 2.7 m on a pile of mass 56 kg and drives it 10 cm into the ground. Determine **10**

i) The velocity with which the weight strikes the pile.

ii) The common velocity with which the weight and the pile begin to move.

iii) Average resistance which brings the weight and the pile to the rest.

32. A block of weight 300 N slides down the curved path from A to B in the vertical plane as shown in the figure. If the block has a velocity of 1.2 m/s down the inclined plane at point A and 8 m/s at B. Determine the work done against friction during the motion from A to B. **10**



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